The Midnight Conservatory

STUDENT ASSIGNMENT

01 Life Processes

Name _____ Batch _____ Date _____

Use the space below each question. Label diagrams and show your working.

01 CBSE 2023 · 31/1/1 · Q8 — EXPLAIN

Bread mould feeds without swallowing pieces of bread. Describe how it obtains soluble nutrients.

02 CBSE 2023 · 31/1/1 · Q9 — EXPLAIN

A section of small intestine has badly damaged villi. Predict the effect on nutrient uptake and give a reason.

03 CBSE 2023 · 31/1/1 · Q23a — EXPLAIN

Trace deoxygenated blood from the body to the lungs. Include the vena cava, heart chambers and pulmonary artery.

04 CBSE 2023 · 31/1/1 · Q23b — EXPLAIN

Compare the final products of glucose breakdown in well-oxygenated human cells and in working muscles short of oxygen.

05 CBSE 2023 · 31/1/1 · Q24 EXPLAIN

Locate nephrons and explain what happens to useful materials in the fluid they initially filter.

06 CBSE 2023 · 31/1/1 · Q29a EXPLAIN

Why is ATP called an energy carrier? Give one use of ATP in a cell.

07 CBSE 2023 · 31/1/1 · Q29b EXPLAIN

How can desert plants acquire carbon dioxide for daytime photosynthesis while limiting water loss?

08 CBSE 2024 · 31/1/1 · Q11 EXPLAIN

What happens to guard cells and the stomatal pore when water leaves the guard cells?

EXTRA WORKING SPACE

09 CBSE 2024 · 31/1/1 · Q22 EXPLAIN

Explain two ways hydrochloric acid helps in the stomach.

10 CBSE 2024 · 31/1/1 · Q29 EXPLAIN

Why do alveoli have thin walls and a rich supply of capillaries?

11 CBSE 2024 · 31/1/1 · Q29 EXPLAIN

Explain how cartilage rings in the trachea and residual air in the lungs help breathing.

12 CBSE 2025 · 31/1/1 · Q8 EXPLAIN

Chewing bread for a while can make it taste sweeter. Name the relevant enzyme and explain the change.

EXTRA WORKING SPACE

13 CBSE 2025 · 31/1/1 · Q23a — EXPLAIN

How does clotting after a cut help maintain circulation? Name the blood component that helps initiate clotting.

14 CBSE 2025 · 31/1/1 · Q23b — EXPLAIN

A root is storing sugars while a young shoot is growing. Which tissue can supply both? Explain why movement may be in different directions.

15 CBSE 2025 · 31/1/1 · Q38 — SHOW WORKING

Write the balanced overall equation for photosynthesis. If 12 molecules of CO_2 are fixed according to the equation, how many molecules of glucose and oxygen form?

16 CBSE 2025 · 31/1/1 · Q38 — EXPLAIN

One plant is shaded. Another is in bright light but its stomata are blocked. Identify the likely limiting input in each.

EXTRA WORKING SPACE

17 CBSE 2026 · 31/1/1 · Q1 QUICK RESPONSE

Name the main route by which carbon dioxide enters a leaf and identify the cells that regulate it.

18 CBSE 2026 · 31/1/1 · Q5 EXPLAIN

Name the pancreatic enzymes that act on proteins and fats. State how bile assists the latter process.

19 CBSE 2026 · 31/1/1 · Q9 EXPLAIN

Why is haemoglobin needed even though some oxygen can dissolve directly in blood plasma?

20 CBSE 2026 · 31/1/1 · Q10 EXPLAIN

Explain how contracting the diaphragm brings air into the lungs. Use changes in volume and pressure.

EXTRA WORKING SPACE

02 Control and Coordination

21 CBSE 2023 · 31/1/1 · Q11 EXPLAIN

A root grows towards a moist region on one side. Identify the response and explain why it is a tropism.

22 CBSE 2023 · 31/1/1 · Q11 EXPLAIN

A seedling is laid horizontally. State the usual response of the root and shoot to gravity.

23 CBSE 2023 · 31/1/1 · Q22 EXPLAIN

A person's blood glucose rises after a meal. Name the hormone released, its source and its effect.

24 CBSE 2023 · 31/1/1 · Q22 EXPLAIN

What normally happens to insulin secretion after blood glucose falls towards normal? Explain the feedback.

EXTRA WORKING SPACE

25 CBSE 2024 · 31/1/1 · Q10 EXPLAIN

At an axon terminal, explain how a signal crosses a typical chemical synapse.

26 CBSE 2024 · 31/1/1 · Q23a EXPLAIN

Compare folding of Mimosa leaves on touch with downward growth of roots. Use direction and growth in your comparison.

27 CBSE 2024 · 31/1/1 · Q23b EXPLAIN

Name the hormone that regulates the metabolism of carbohydrates, proteins and fats. Which element is needed to make it?

28 CBSE 2024 · 31/1/1 · Q23b EXPLAIN

Why can a long-term shortage of iodine lead to an enlarged thyroid?

EXTRA WORKING SPACE

29 CBSE 2024 · 31/1/1 · Q30 DIAGRAM

Use a flow chart to trace the reflex response when a finger touches a hot object.

30 CBSE 2024 · 31/1/1 · Q30 EXPLAIN

Why can someone withdraw a hand before consciously recognising pain?

31 CBSE 2025 · 31/1/1 · Q9 EXPLAIN

Predict how a shoot bends when light comes from the left. Explain the role of auxin.

32 CBSE 2025 · 31/1/1 · Q9 EXPLAIN

A seedling receives equal light from both sides. Predict the effect on its lateral bending due to light.

EXTRA WORKING SPACE

33 CBSE 2025 · 31/1/1 · Q29 EXPLAIN

How does a touched Mimosa leaflet fold without using muscles or increasing in length?

34 CBSE 2025 · 31/1/1 · Q29 EXPLAIN

Why does a pea tendril coil around a support after contact? Name the response.

35 CBSE 2026 · 31/1/1 · Q4 QUICK RESPONSE

Which part of the brain helps someone walk along a narrow beam without losing balance?

36 CBSE 2026 · 31/1/1 · Q4 EXPLAIN

A person decides to pick up a pencil and then grips it smoothly. Distinguish the main roles of the cerebrum and cerebellum.

EXTRA WORKING SPACE

37 CBSE 2026 · 31/1/1 · Q8 EXPLAIN

Why is an automatic blink in response to a sudden threat described as a reflex? Give two characteristics.

38 CBSE 2026 · 31/1/1 · Q11a EXPLAIN

Classify starting to chew and salivating at the sight of food as voluntary or involuntary. Explain the distinction.

39 CBSE 2026 · 31/1/1 · Q14 EXPLAIN

After a skin receptor is stimulated, which neuron carries the signal towards the central nervous system? Which carries it towards a muscle?

40 CBSE 2026 · 31/1/1 · Q14 EXPLAIN

Compare the functions of a neuron and a nephron. Name the system associated with each.

EXTRA WORKING SPACE

03 How Do Organisms Reproduce?

41 CBSE 2022 · 31/1/1 · Q3 EXPLAIN

Mustard and hibiscus can self-pollinate within a flower, while a pistillate papaya flower cannot. Explain.

42 CBSE 2022 · 31/1/1 · Q4 EXPLAIN

Distinguish fragmentation in Spirogyra from regeneration in Planaria when part of each is separated.

43 CBSE 2022 · 31/1/1 · Q10 EXPLAIN

State the role of the testes, scrotum and vas deferens in human reproduction.

44 CBSE 2022 · 31/1/1 · Q10 EXPLAIN

Compare human sperm and ovum in motility and cytoplasm.

EXTRA WORKING SPACE

45 CBSE 2023 · 31/1/1 · Q12 EXPLAIN

Identify the vegetative structures from which new potato and Bryophyllum plants can grow.

46 CBSE 2023 · 31/1/1 · Q35 EXPLAIN

What happens to a flower’s ovary and ovules after successful fertilisation?

47 CBSE 2023 · 31/1/1 · Q35 EXPLAIN

Identify two parts of a typical bisexual flower that do not directly produce gametes and state one role for each.

48 CBSE 2024 · 31/1/1 · Q16 EXPLAIN

Name an organism that reproduces by multiple fission and distinguish it from binary fission.

EXTRA WORKING SPACE

49 CBSE 2024 · 31/1/1 · Q35a EXPLAIN

Compare an intrauterine device and hormonal pills by where or how each acts to prevent pregnancy.

50 CBSE 2024 · 31/1/1 · Q35b EXPLAIN

Explain why Rhizopus spores may survive on dry bread yet fail to multiply there.

51 CBSE 2025 · 31/1/1 · Q22 EXPLAIN

Describe three stages in Hydra budding, from the start of the bud to an independent animal.

52 CBSE 2025 · 31/1/1 · Q35a EXPLAIN

Give one reproductive function each of the ovary, oviduct and uterus.

EXTRA WORKING SPACE

53 CBSE 2025 · 31/1/1 · Q35a EXPLAIN

Compare a condom with vasectomy in how each blocks the path of sperm. Which also reduces transmission of many STIs?

54 CBSE 2025 · 31/1/1 · Q35b EXPLAIN

Contrast self-pollination and cross-pollination using the source of pollen. Which can bring together alleles from two parent plants?

55 CBSE 2026 · 31/1/1 · Q2 EXPLAIN

One single-celled organism divides into two after duplicating its genetic material. Name the process and outline the sequence.

56 CBSE 2026 · 31/1/1 · Q11b EXPLAIN

Why can a flower be pollinated without being fertilised?

EXTRA WORKING SPACE

57 CBSE 2026 · 31/1/1 · Q12 DIAGRAM

Draw a simple pistil after pollination. Label stigma, style, ovary, ovule and the pollen tube as it grows towards the ovule.

58 CBSE 2026 · 31/1/1 · Q16a EXPLAIN

What happens after sperm and ovum fuse in the oviduct? Include the first cell formed and its eventual destination.

59 CBSE 2026 · 31/1/1 · Q16a EXPLAIN

If fertilisation does not occur, what happens to the thickened uterine lining and why?

60 CBSE 2026 · 31/1/1 · Q16b EXPLAIN

What happens to a bread mould sporangium when mature, and how does this aid reproduction?

EXTRA WORKING SPACE

04 Heredity

61 CBSE 2022 · 31/1/1 · Q5b — SHOW WORKING

Pure violet-flowered peas are crossed with pure white-flowered peas. If violet is dominant, find the expected number of white plants among 160 F_2 offspring.

62 CBSE 2022 · 31/1/1 · Q5b — EXPLAIN

Could a violet F_2 plant carry the allele for white flowers? Explain using its possible genotypes.

63 CBSE 2022 · 31/1/1 · Q14 — EXPLAIN

State the number of autosomes and sex chromosomes in a typical human body cell and in a normal ovum.

64 CBSE 2022 · 31/1/1 · Q14 — DIAGRAM

Draw a simple flow chart showing the two possible sex chromosome outcomes when an X-bearing ovum is fertilised.

EXTRA WORKING SPACE

65 CBSE 2022 · 31/1/1 · Q14 EXPLAIN

Explain why sex determination in some reptiles is different from the usual XX/XY model in humans.

66 CBSE 2023 · 31/1/1 · Q10 EXPLAIN

Distinguish a gene from a chromosome and explain their relationship.

67 CBSE 2023 · 31/1/1 · Q18 EXPLAIN

In the usual XX/XY model, from which parent does an XY child receive the X chromosome? Explain.

68 CBSE 2023 · 31/1/1 · Q38 SHOW WORKING

Pure round yellow peas (RRYY) are crossed with wrinkled green peas (rryy). List parental gametes and predict all F_1 genotypes and phenotypes.

EXTRA WORKING SPACE

69 CBSE 2023 · 31/1/1 · Q38 SHOW WORKING

An RrYy plant forms gametes. List the possible combinations and their expected proportions if the genes assort independently.

70 CBSE 2023 · 31/1/1 · Q38 SHOW WORKING

Two RrYy peas are crossed. With independent assortment and full dominance, find the chance of wrinkled green offspring.

71 CBSE 2024 · 31/1/1 · Q8 EXPLAIN

At what stage is the usual XX or XY combination established in humans? Explain the event.

72 CBSE 2024 · 31/1/1 · Q9 SHOW WORKING

A typical human body cell has 46 chromosomes. How many chromosomes occur in each normal gamete, and how many in the zygote?

EXTRA WORKING SPACE

73 CBSE 2024 · 31/1/1 · Q38 EXPLAIN

Tallness (T) is dominant. Design a cross that distinguishes a tall TT pea from a tall Tt pea.

74 CBSE 2025 · 31/1/1 · Q11 SHOW WORKING

Cross round green peas RRyy with wrinkled yellow peas rrYY. Explain why the F_1 phenotype differs from that of both parents.

75 CBSE 2025 · 31/1/1 · Q30a EXPLAIN

Why can DNA in chromosomes account for the transmission of traits from one generation to the next?

76 CBSE 2025 · 31/1/1 · Q30b EXPLAIN

Predict what would happen to chromosome number over generations if gametes were formed without reduction division.

EXTRA WORKING SPACE

77 CBSE 2026 · 31/1/1 · Q6 SHOW WORKING

Assuming the usual equal X- and Y-bearing sperm model and independent births, what is the probability that two children will both have the XX combination?

78 CBSE 2026 · 31/1/1 · Q15a SHOW WORKING

Self a heterozygous tall pea, Tt. Give the genotypic and phenotypic ratios expected in the offspring.

79 CBSE 2026 · 31/1/1 · Q15a SHOW WORKING

Among the tall offspring of Tt × Tt, what fraction are expected to be heterozygous?

80 CBSE 2026 · 31/1/1 · Q16a EXPLAIN

A heat-tolerant variant exists in a bacterial population before the water warms. Explain why it may become common after warming.

EXTRA WORKING SPACE

05 Our Environment

81 CBSE 2022 · 31/1/1 · Q7a — EXPLAIN

Explain how ozone forms in the upper atmosphere and why it matters to living organisms.

82 CBSE 2022 · 31/1/1 · Q7b — EXPLAIN

Why does a typical household aquarium require more deliberate maintenance than an established natural pond?

83 CBSE 2022 · 31/1/1 · Q13 — EXPLAIN

Predict two consequences of losing decomposers from an ecosystem.

84 CBSE 2023 · 31/1/1 · Q26 — EXPLAIN

Explain how a persistent pesticide can reach high concentrations in fish-eating birds even when its concentration in the water is low.

EXTRA WORKING SPACE

85 CBSE 2023 · 31/1/1 · Q33a — EXPLAIN

Design an indoor freshwater ecosystem with producers and decomposers. Explain why light and a balanced population matter.

86 CBSE 2023 · 31/1/1 · Q33b — SHOW WORKING

In plants → rat → snake → hawk, rats receive 20,000 J. Estimate the energy transferred to snakes and hawks using the 10% rule.

87 CBSE 2024 · 31/1/1 · Q15 — EXPLAIN

A proposed chain reads grass → snake → hawk. Identify the missing trophic role and insert a suitable organism.

88 CBSE 2024 · 31/1/1 · Q18 — EXPLAIN

A pesticide is applied to crops. Why does knowing that it kills pests fail to explain why a top predator accumulates it?

EXTRA WORKING SPACE

89 CBSE 2024 · 31/1/1 · Q33a — EXPLAIN

Predict the initial effect of removing deer from a plants → deer → lion chain. How could a food web alter the lion’s outcome?

90 CBSE 2024 · 31/1/1 · Q33b — EXPLAIN

Name the class of chemicals historically linked to ozone depletion and the international agreement that controls many of them.

91 CBSE 2025 · 31/1/1 · Q15 — EXPLAIN

Why is a chain grass → snake → frog biologically implausible? Supply a corrected four-level terrestrial example.

92 CBSE 2025 · 31/1/1 · Q16 — SHOW WORKING

Green leaves receive 80,000 J of incident sunlight. If 1% becomes stored food energy, calculate the energy captured and the energy not captured.

EXTRA WORKING SPACE

93 CBSE 2025 · 31/1/1 · Q20 EXPLAIN

Explain why describing how ozone forms does not explain the large observed decline in stratospheric ozone.

94 CBSE 2025 · 31/1/1 · Q26 EXPLAIN

A heavy rain follows pesticide application. Explain one effect on a non-living part and one effect on a living part of a nearby pond.

95 CBSE 2026 · 31/1/1 · Q7 EXPLAIN

Sort a banana peel, a polythene bag, paper and a plastic pen barrel as biodegradable or non-biodegradable in ordinary disposal conditions.

96 CBSE 2026 · 31/1/1 · Q13 SHOW WORKING

Producers store 40,000 J. Apply the 10% rule to estimate the energy at the primary, secondary and tertiary consumer levels.

EXTRA WORKING SPACE

97 CBSE 2026 · 31/1/1 · Q13 DIAGRAM

Draw a four-level energy pyramid for grass, grasshopper, frog and snake. Put the organisms in order from base to top and explain the broad base.

98 CBSE 2026 · 31/1/1 · Q13 SHOW WORKING

The secondary consumers receive 120 J in a food chain. Estimate the energy at the producer level, assuming 10% transfer at each step.

99 CBSE 2026 · 31/1/1 · Q14 EXPLAIN

Contrast the roles of a consumer and a decomposer. Explain which one returns mineral nutrients from dead material.

100 CBSE 2023 · 31/1/1 · Q33b SHOW WORKING

In a four-level chain, the top consumer receives 25 J. Under the 10% rule, estimate how much energy the primary consumer level must receive.

EXTRA WORKING SPACE

The reference shelf

Each citation identifies the source idea in a CBSE Class 10 Science board paper. Prompts are paraphrased, split into individual skills, or extended with fresh data. Question labels refer to the source paper, and worksheet numbers refer to this assignment.

2022 · Science 086 · Paper 31/1/1 · Term II

Open question paper reference

2023 · Science 086 · Paper 31/1/1

Open question paper reference

2024 · Science 086 · Paper 31/1/1

Open question paper reference

2025 · Science 086 · Paper 31/1/1

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2026 · Science 086 · Paper 31/1/1

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